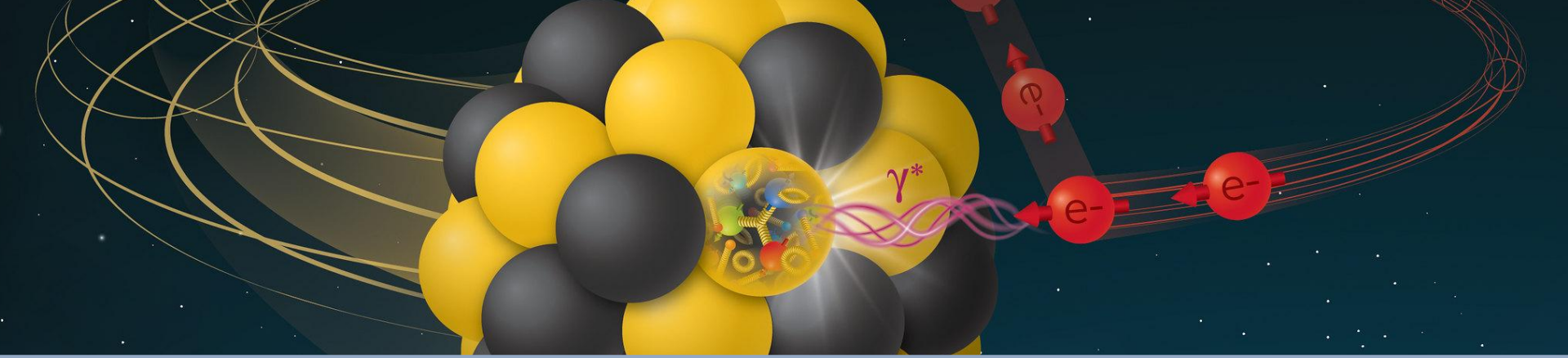


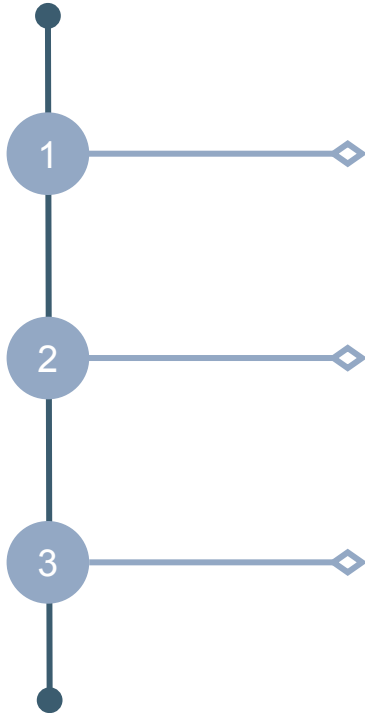


# Dependence of $D^0$ topological variables on $Q^2$ , $p_T$ , and $y$ ( $\eta$ )

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# Introduction

# Electron-Ion Collider (EIC)

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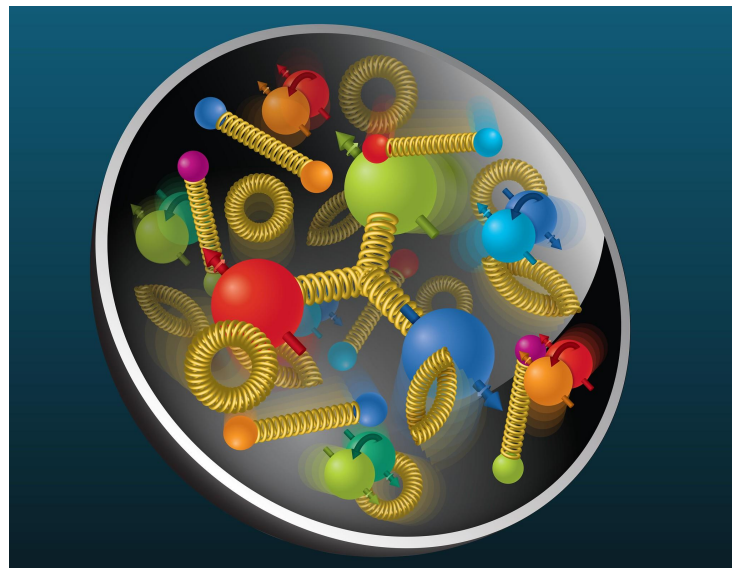
Image 3D distributions of gluons and sea quarks (99% of proton mass)

Solve proton spin puzzle

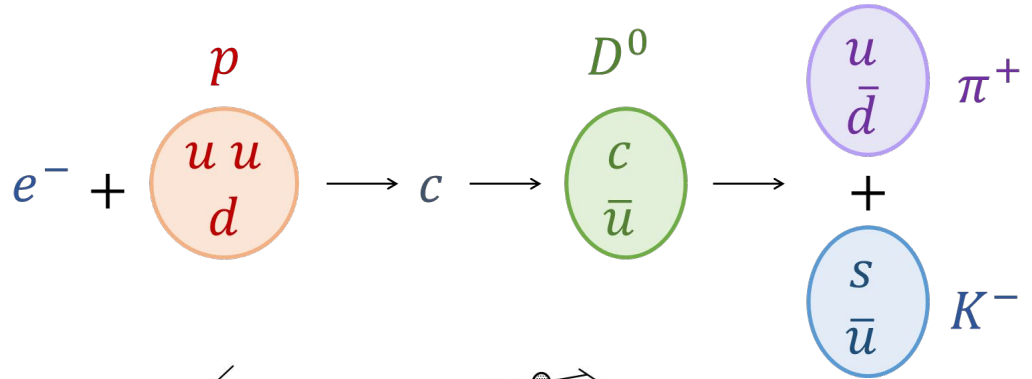
Search for color glass condensate

Understand confinement

Study quark and gluon distributions in nuclei



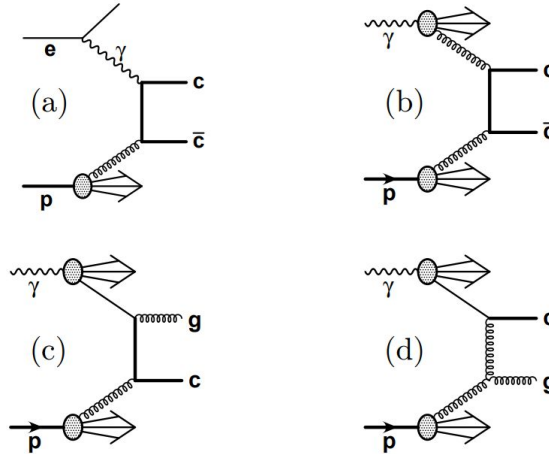
# Studying $D^0$



Charm quarks as probes

Analyze  $D^0$  mesons

Topological variables to distinguish  
signal from background



# $D^0$ topological variables

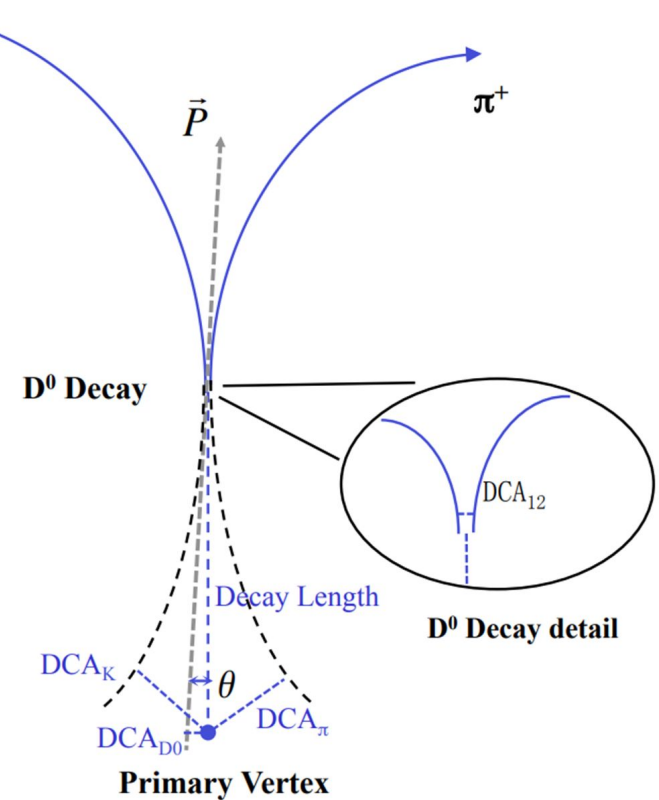
$DCA_{\pi}$  and  $DCA_K$  : distance of pion/kaon to primary vertex

$DCA_{12}$  : distance between pion and kaon pair

$DCA_{D^0}$  : distance of parent particle to primary vertex

Decay length  $L$ : distance from primary to secondary vertex

$\theta$ : angle between  $D^0$  momentum and vertex vector



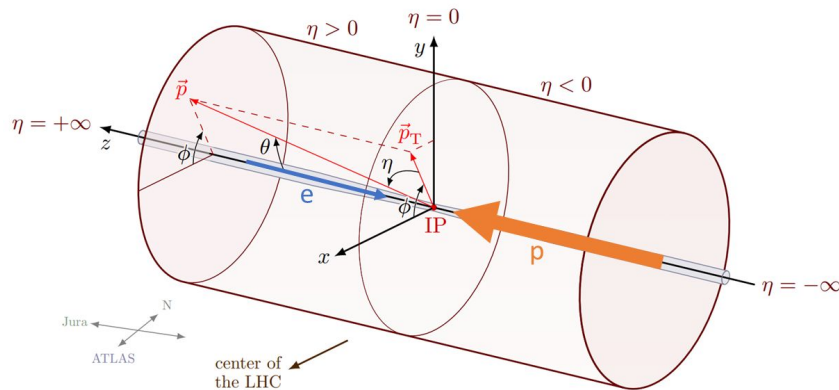
# Goals

**Overall:** Optimize  $D^0$  topological cuts using machine learning.

**This semester:** Analyze dependences of topological variables on  $Q^2$ ,  $p_T$ , and  $y$  ( $\eta$ ).

Compare

- $Q^2 > 1$  vs.  $Q^2 > 100$  samples
- $p_T$  and  $y$  slices
- Using reconstructed (RC) vs. Monte Carlo (MC) primary vertex



# Data samples and signal selection

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## Data samples

- PYTHIA8, ep,  $Q^2 > 1$  and  $Q^2 > 100$
- 50 runs for each  $Q^2$
- ~26500 total events each

(JLab directories)

## Signal selection

From all pions and kaons:

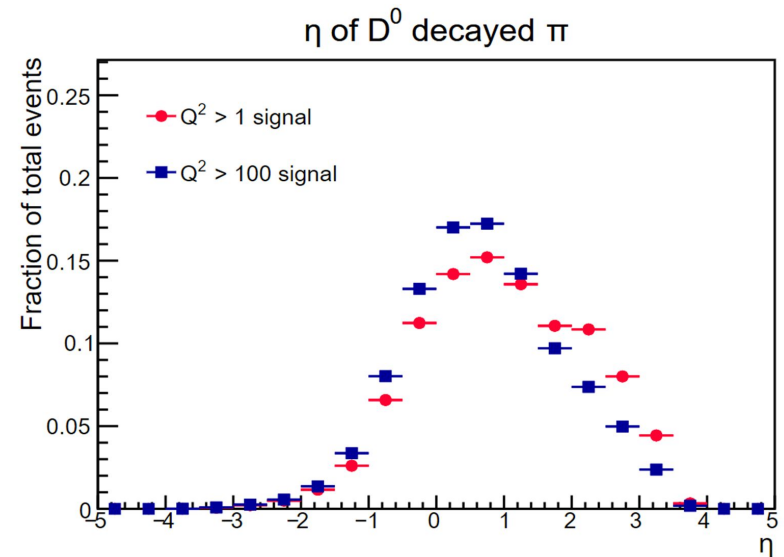
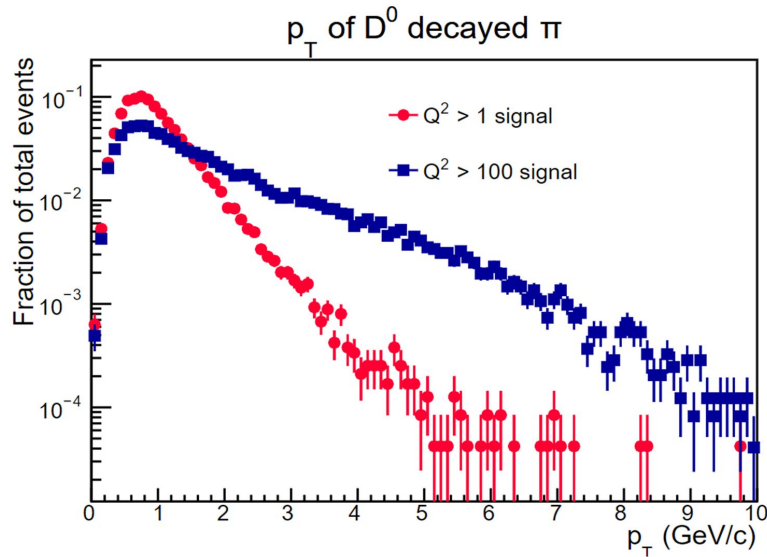
- $\text{dist} > 10^{-4}$  mm from primary vertex (non-primary)
- $\text{DCA}_\pi, \text{DCA}_K > 0.02$  mm
- $D^0$  parent
- $\pi K$  pair with opposite charges

([github.com/marrbnl/ePIC/tree/main/HF\\_rec](https://github.com/marrbnl/ePIC/tree/main/HF_rec)  
[o/helix/analysis.cxx](#))



# $p_T$ and $y(\eta)$ distributions

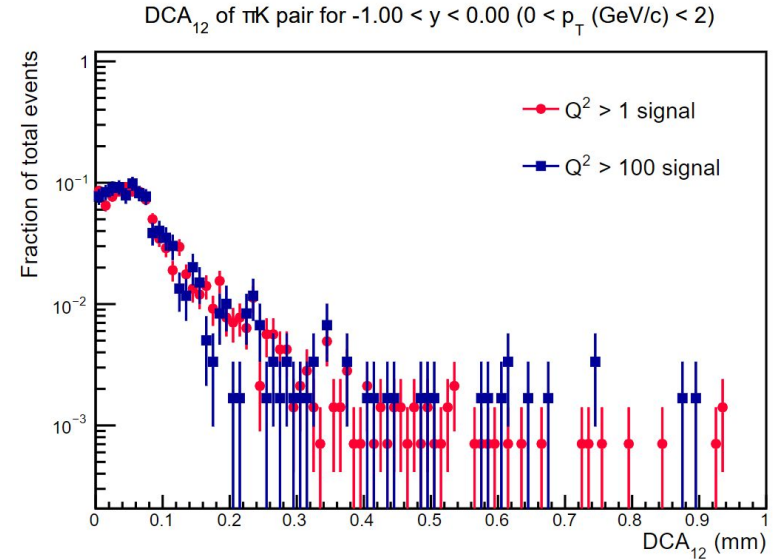
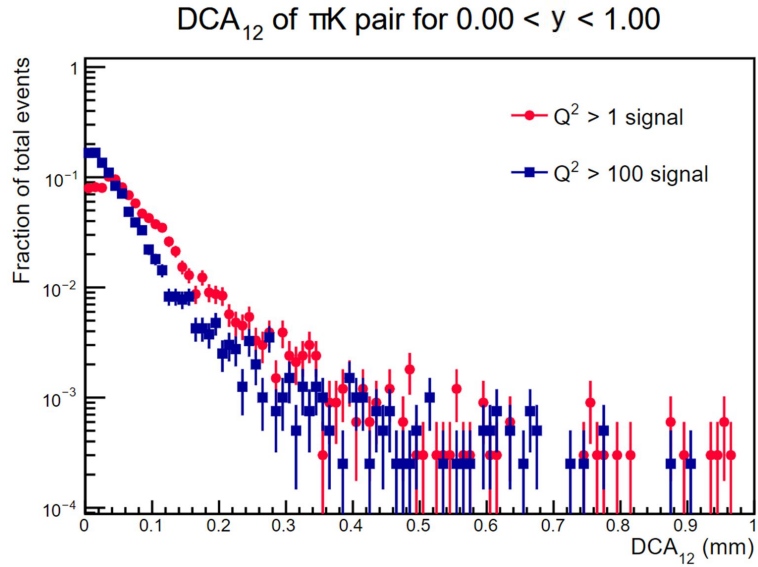
# Total $p_T$ and $\eta$



Higher  $Q^2$  results in higher average  $p_T$ .

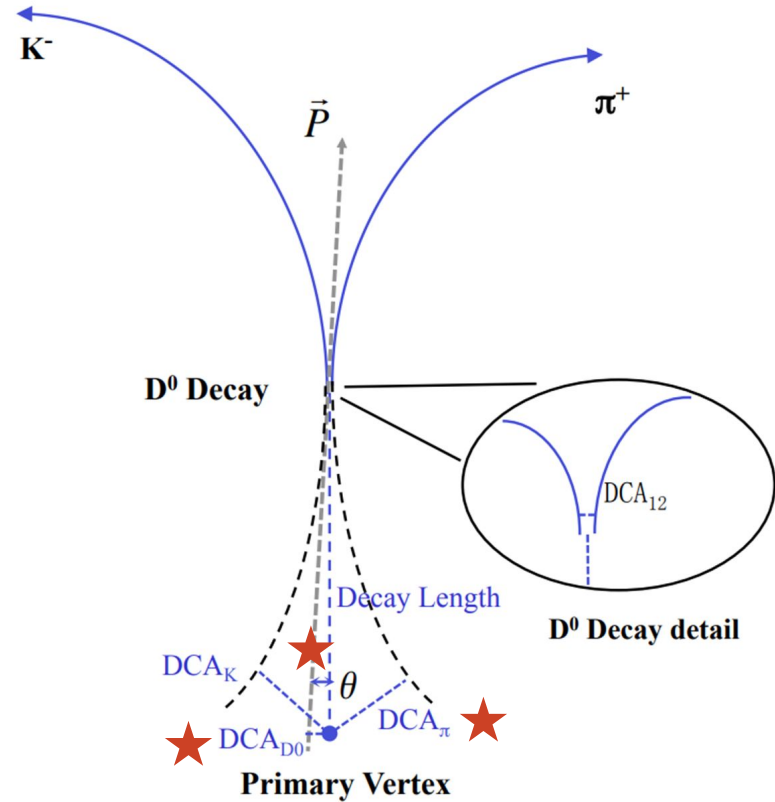
More particles in forward  $\eta$  due to asymmetric ep collision.

# Correlation of $p_T$ and $Q^2$ dependence

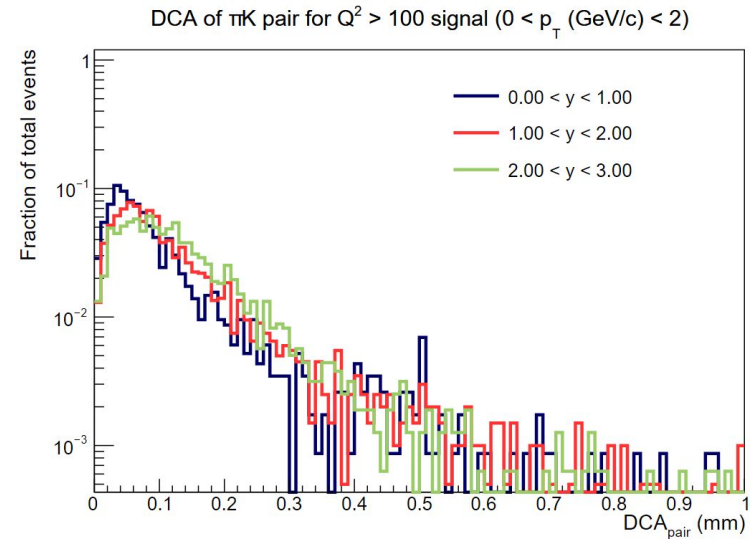
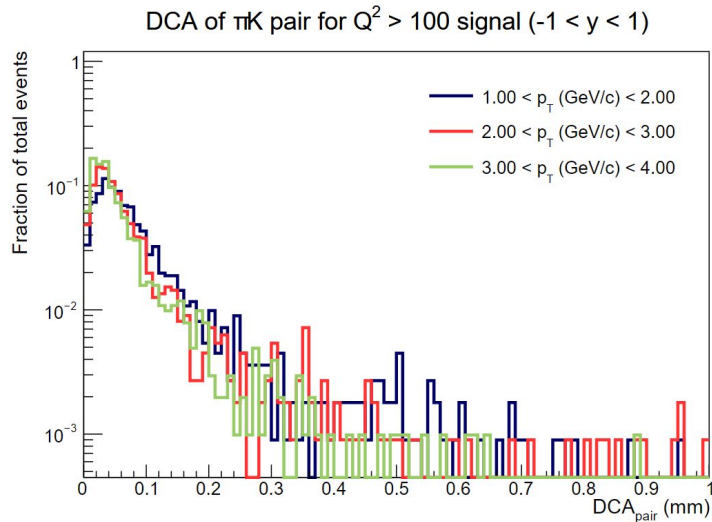


Restricting  $p_T$  removes  $Q^2$  dependence.

# Topological variable dependences on $p_T$ and $y$



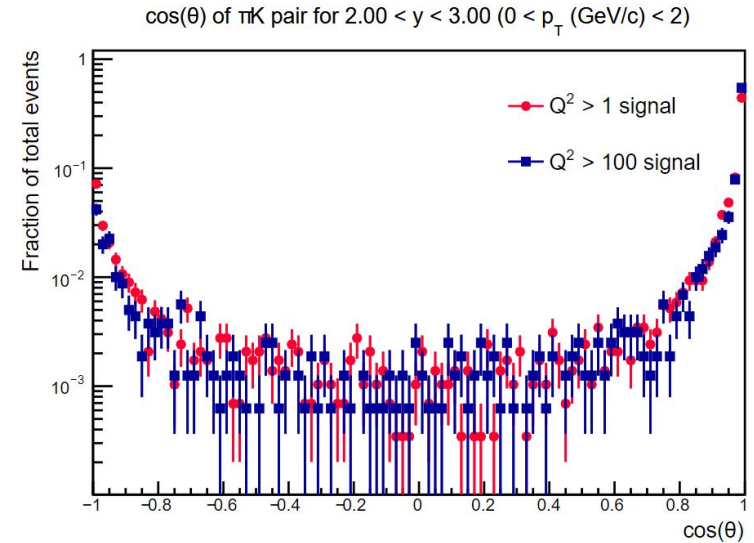
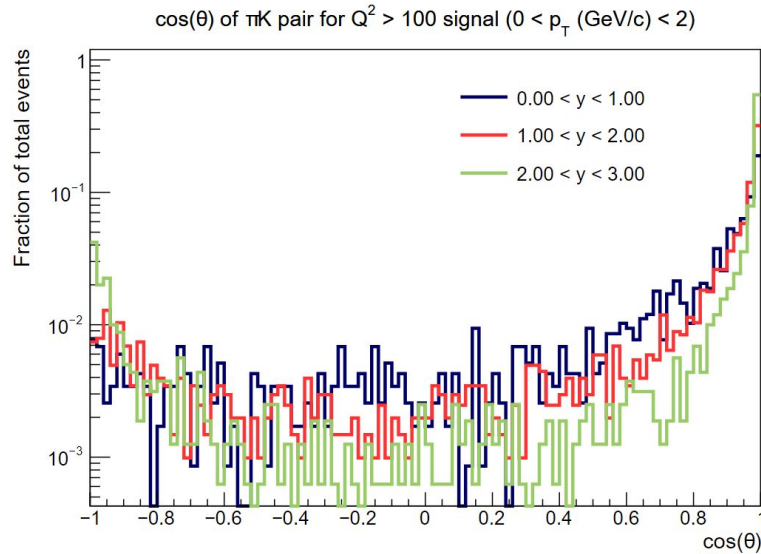
# $p_T$ and $y$ dependences of $DCA_{D0}$



Smallest average  $DCA_{D0}$  at high  $p_T$  and mid  $y$ .

Best track resolution in these ranges.

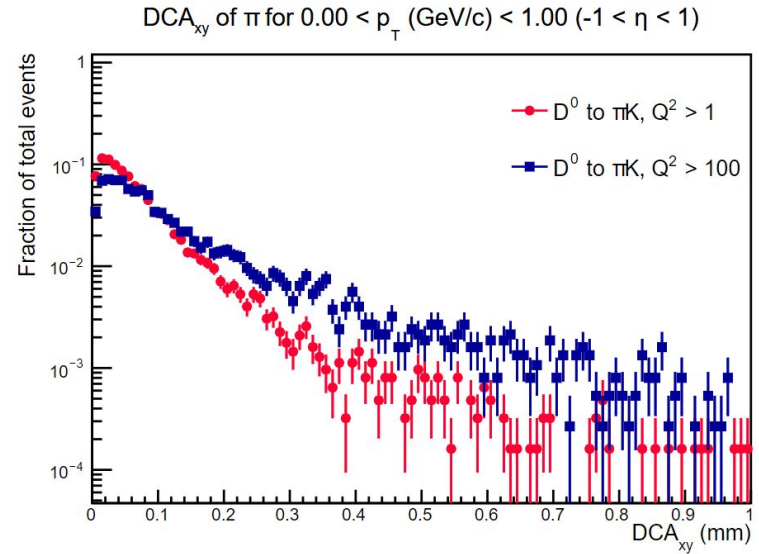
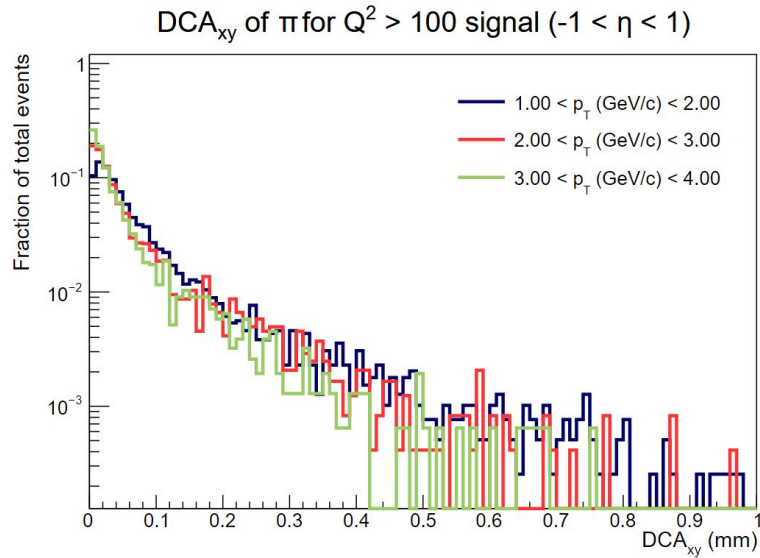
# y dependence of $\cos(\theta)$



Significant fraction of candidates near  $\cos(\theta) = -1$  in forward y.

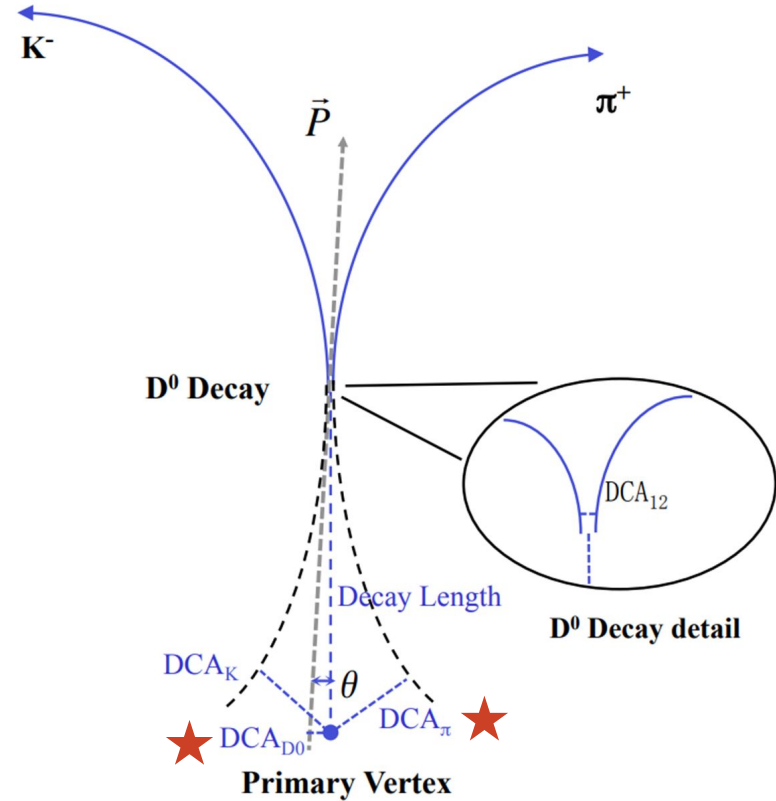
Poor track resolution in this y range.

# $Q^2$ dependence of $DCA_{\pi\pi}$



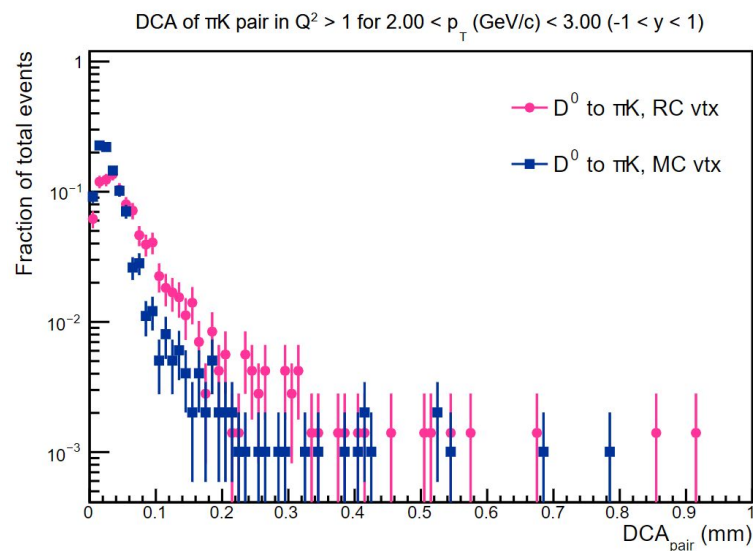
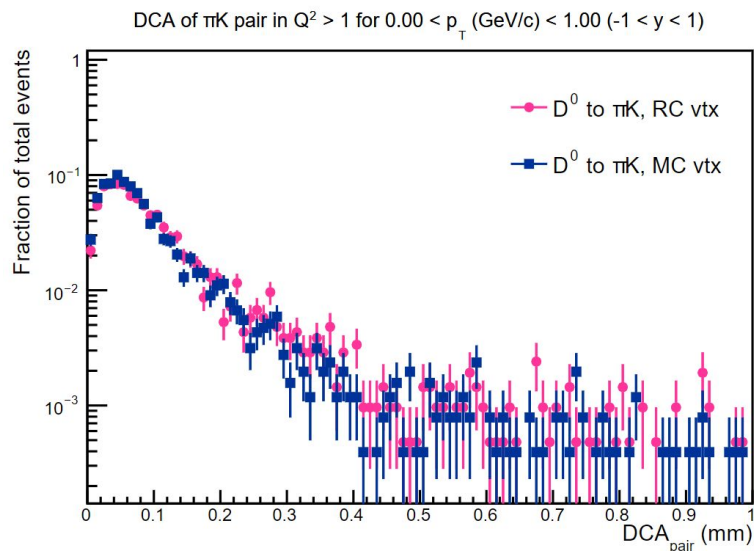
$Q^2$  dependence in opposite direction as  $p_T$  dependence.

# Effects of primary vertex reconstruction





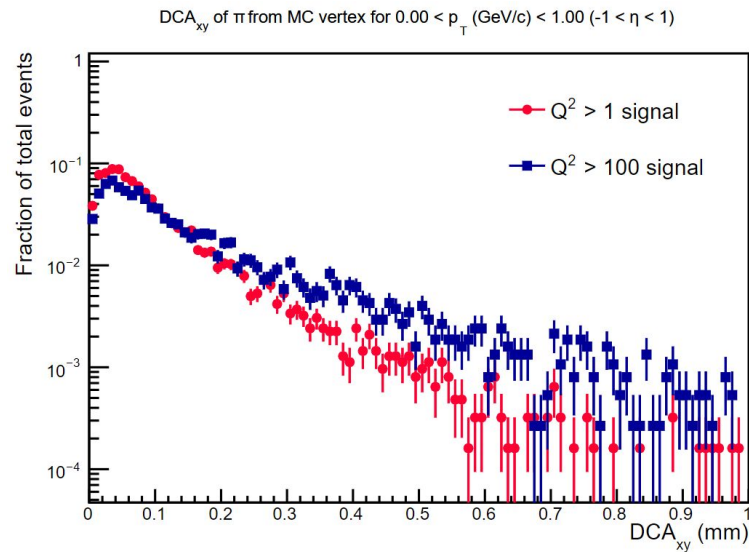
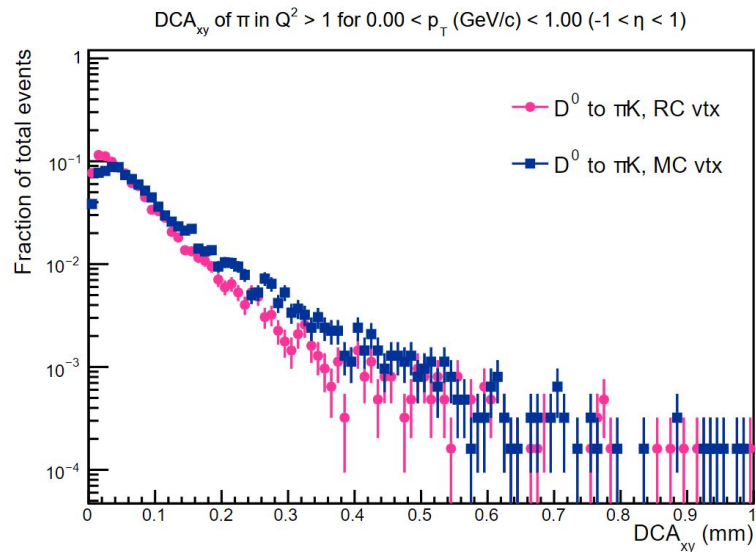
# MC vs. RC primary vertex for $DCA_{D^0}$



RC is wider due to limited primary vertex resolution.

Greatest effect at high  $p_T$  and mid  $y$ , where track resolution is best.

# MC vs. RC primary vertex for $DCA_{\pi}$



RC is narrower.

$Q^2$  dependence still present in MC distributions.

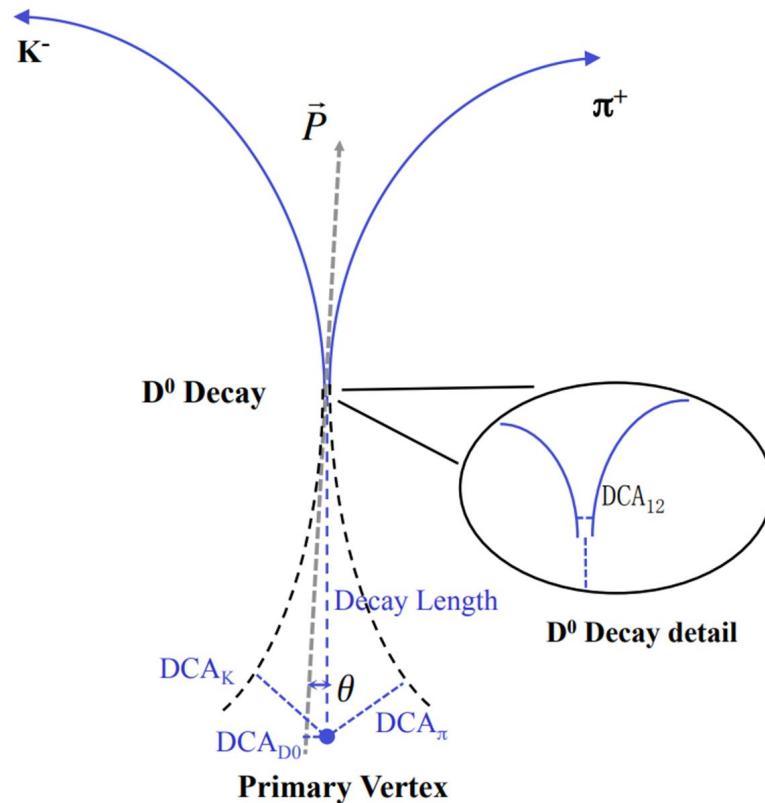
# Conclusion

# Summary

Strong  $p_T$  and  $y$  ( $\eta$ ) dependence for all topological variables.

$Q^2$  dependence correlated with  $p_T$  dependence, except for  $DCA_{\pi/K}$ .

Reconstructed primary vertex distributions are wider, except for  $DCA_{\pi/K}$  (and L).

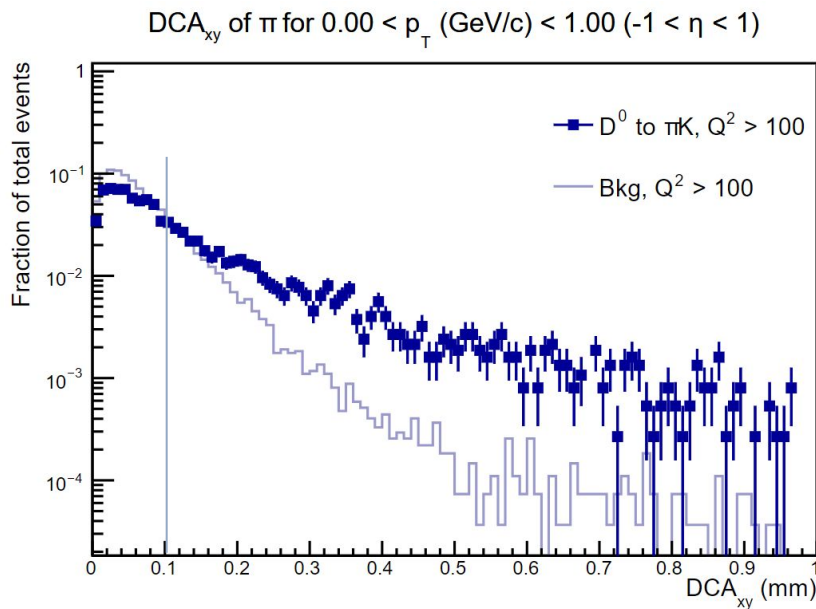


# Outlook

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Analyze background distributions.

Use machine learning to optimize topological cuts for maximum signal : background ratio.



# References

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Phys. Rev. C 99, 034908 (2019)

<https://www.bnl.gov/eic/>

[https://github.com/marrbnl/ePIC/blob/main/HF\\_reco/helix/](https://github.com/marrbnl/ePIC/blob/main/HF_reco/helix/)

Thank you